CIS 446/CIS 546 Secure Software Development

Assignment 1 (50 points)

Due: 11:59pm on March 3rd, 2026

This is an individual assignment.

The source code of SecureED can be found at: https://github.com/onniegit/SecureEd

(Task 1) (20 points) Apply input validation to login.php (located in /app/src) to satisfy the following

requirements:

- the username is a valid email address in the format of username@domain.com
- the password is a valid text with the following constraints
  - length between 8 to 16 (inclusive)
  - consists of characters a-z, A-Z, and 0-9 with at least one capital letter and one number

[hint: Regular expressions can help pattern match. PHP provides functions for regular expressions, for example,

https://www.php.net/manual/en/function.preg-match.php

A tutorial on regular expression from W3shools can be found at:

https://www.w3schools.com/php/php_regex.asp]

Please provide the screenshot of a general error message when error occurs.

(Task 2) (20 points) Mitigate the SQLi vulnerability in code of login.php.

[Several mitigation strategies are available.

For example, parameterize SQL queries:

https://www.php.net/manual/en/pdo.prepared-statements.php

SecureED uses SQLITE 3. Reference to SQLite3Stmt class can be found at:

https://www.php.net/manual/en/book.sqlite3.php

PHP Data Objects:

https://www.php.net/manual/en/book.pdo.php]

Please provide the screenshot of a general error message without exposing the database information when error occurs.

Deliverables:

(1) (10 Points) Prepare a report which should describe SQL Injection vulnerability in detail with proper example. Include the screenshot of error message from Task 1 and Task 2 in the report.
(2) Submit the modified source code (login.php) and the report on Canvas by 11:59pm on March 3rd, 2026.